

Research Ethics and Integrity

120 minutes · 6 self-check questions

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The principles governing research with human participants, how ethical review actually works, and the integrity questions — authorship, plagiarism, data fabrication and the use of AI — that decide whether your work stands.

Learning objectives

- State the four principles of research ethics and apply them to a scenario
 - Explain what makes consent genuinely informed, voluntary and documented
 - Assemble an ethics application and anticipate what a committee will query
 - Distinguish anonymity from confidentiality and implement both
 - Identify vulnerable groups and the additional safeguards they require
 - Apply authorship criteria and recognise research misconduct
 - State a defensible position on the use of AI tools in research
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Principles and their origins

- The Nuremberg Code followed the Nazi medical experiments; the Declaration of Helsinki and the Belmont Report followed later abuses, including Tuskegee
 - Autonomy — people decide for themselves whether to take part, having understood what is involved
 - Beneficence — maximise benefit; Non-maleficence — minimise harm; the two are weighed together
 - Justice — burdens and benefits are distributed fairly, and no group is used as a convenient subject pool for research that will benefit others
 - Justice has particular weight in African research settings, where communities have historically borne research burden without sharing in benefit
 - Ethical conduct is not a form to be completed; it is a continuing obligation through the study
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Informed consent

- Information — purpose, procedures, duration, risks, benefits, alternatives, confidentiality limits, and who to contact
- Comprehension — in the participant's language, at an appropriate reading level, with opportunity to ask questions

- Voluntariness — free from coercion or undue inducement; compensation should cover cost and time, not act as an incentive that overrides judgement
- Documentation — signature, thumbprint or witnessed verbal consent, as appropriate to context and literacy
- The right to withdraw at any time, without giving a reason and without penalty, must be stated and honoured
- Children give assent while a parent or guardian gives consent; the age of assent varies by jurisdiction
- Consent is a process, not a signature: it should be revisited in longitudinal and ethnographic work
- Waiver of consent may be granted for anonymised secondary data or minimal-risk record review, but only by the committee

IN PRACTICE

Where a clinician recruits their own patients, or a lecturer their own students, the power relationship can make refusal feel impossible. Address this explicitly in the application — usually by having someone independent take consent.

Review and approval

- Institutional Review Boards and Research Ethics Committees review before any data are collected
- In Ghana, approvals commonly involve institutional RECs and, depending on the work, the Ghana Health Service Ethics Review Committee or the FDA
- A typical application requires the protocol, instruments, information sheet, consent form, translations, CVs and a data management plan
- Committees most often query: unclear risk-benefit, weak consent for vulnerable groups, inadequate data protection, undisclosed conflicts, and underpowered designs
- Approval is time-limited; amendments and extensions must be submitted, and adverse events reported
- Publishing without approval is a serious breach — many journals require the approval number, and retraction follows if it was never obtained

Confidentiality and data protection

- Anonymity means even the researcher cannot link data to a person; confidentiality means the researcher can but will not disclose
- Most studies offer confidentiality, not anonymity — say which, accurately
- De-identify at the earliest point; keep any linking key in a separate, encrypted, access-controlled file
- Ghana's Data Protection Act 2012 (Act 843) governs personal data, requiring lawful basis, purpose limitation and security

- State retention periods and how data will be destroyed or archived
 - Qualitative data are especially re-identifiable — a role plus a district can identify someone even without a name
 - Confidentiality has limits: disclosure of imminent harm or of certain crimes may compel action, and participants must be told this in advance
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Integrity, authorship and AI

- Fabrication (inventing data), falsification (altering data) and plagiarism are the three classical forms of misconduct
- Self-plagiarism and duplicate publication are also breaches, even though the text is your own
- ICMJE authorship requires all four: substantial contribution, drafting or critical revision, final approval, and accountability for the work
- Supervisors are not automatically authors; funders and heads of department are not authors by virtue of position — gift authorship is a breach
- Agree the authorship order in writing at the start, not after the results arrive
- Declare funding and conflicts of interest; register trials prospectively and consider preregistering observational protocols
- Report negative and null results — suppressing them distorts the literature and wastes participants' contribution
- AI tools may assist with language, structure and code, but the researcher remains accountable for every claim; AI cannot be an author, must not be used to generate data or references, and its use should be disclosed as journals and institutions increasingly require
- Never paste identifiable participant data into a public AI service — that is a data protection breach regardless of the research value

Worked examples

Consent when the researcher holds power over the participant

A lecturer wants to survey her own final-year students about their experience of the programme. She will administer it herself, at the end of a lecture she teaches.

1. Students may reasonably fear that declining, or answering critically, could affect their marks.
2. That fear compromises voluntariness even if the lecturer would never act on it — the perception is enough.
3. Fix one: an independent colleague, with no assessment role, administers and holds the data.
4. Fix two: collect responses anonymously, and only after final marks are released.
5. The information sheet must state plainly that participation has no bearing on assessment.
6. Name the power relationship explicitly in the ethics application. Committees notice when it is left out.

THE POINT

Whenever the researcher can affect the participant's grades, treatment or employment, address it in the design. Saying "participation was voluntary" does not make it so.

How a quotation identifies someone with no name attached

A report on medical errors quotes: "As the only male theatre nurse at the district hospital, I felt I could not challenge the surgeon."

1. No name appears, yet anyone in that district can identify the speaker immediately.
2. Role, gender, setting and seniority combine into an identifier even when each alone would not.
3. The consequences here are real — this nurse could face professional retaliation.
4. Options: remove the role detail, aggregate the setting to "a district hospital in the region", or paraphrase rather than quote.
5. Where the detail is essential to the point, go back to the participant and ask before publishing.
6. Say in the information sheet how quotations will be handled, so consent covers what you actually do.

THE POINT

Anonymisation means removing what identifies, not just removing names. Read every quotation as someone in that workplace would read it.

Key terms

Autonomy

The principle that competent people decide for themselves whether to participate.

Beneficence

The obligation to maximise benefit and minimise harm to participants.

Justice

Fair distribution of the burdens and benefits of research across groups.

Assent

A child's own agreement to take part, alongside a guardian's consent.

Anonymity

Data cannot be linked to an individual by anyone, including the researcher.

Confidentiality

Identifiable data exist but access is restricted and disclosure prevented.

Gift authorship

Naming as author someone who has not met the criteria for authorship.

Equipoise

Genuine uncertainty about which trial arm is superior, the ethical basis for randomising.

Activity

You plan to interview nurses about errors they have witnessed. Draft the confidentiality paragraph of your information sheet, including its limits. Then list every way a reader might identify a participant from a quotation, and state how you would prevent each.

Further reading

- **World Medical Association — Declaration of Helsinki** — Short, foundational and worth reading in full at least once.
- **The Belmont Report (1979)** — The origin of respect for persons, beneficence and justice.
- **ICMJE Recommendations — authorship criteria** — The four-part test, and what disqualifies a contributor.
- **Ghana Data Protection Act 2012 (Act 843)** — The legal framework for personal data in Ghana.

Self-check questions

1. A researcher offers unemployed participants an amount equal to two weeks' wages to join a study with real risk. The principal concern is:
 - a. The budget is too large
 - b. Undue inducement compromising voluntariness
 - c. A breach of confidentiality
 - d. Nothing — compensation is always acceptable
2. A study records names and phone numbers so participants can be followed up, storing them securely and never disclosing them. This offers:
 - a. Anonymity
 - b. Confidentiality
 - c. Both
 - d. Neither
3. A head of department who provided no intellectual input asks to be listed as an author. Under ICMJE criteria this is:
 - a. Acceptable, as departmental support is a contribution
 - b. Gift authorship, which is a breach of publication ethics
 - c. Acceptable if they are listed last
 - d. Required by most institutions
4. Data collection began before ethical approval was granted, though approval was later obtained. The likely consequence is:
 - a. None, since approval was eventually granted
 - b. The data collected beforehand are unusable and publication may be refused or retracted
 - c. A brief delay only
 - d. The committee will backdate the approval
5. Which use of an AI tool is clearly unacceptable?
 - a. Improving the grammar of a paragraph you wrote
 - b. Generating plausible-looking references you do not verify
 - c. Explaining a statistical concept to you
 - d. Helping debug your analysis code
6. A trial finds the new intervention no better than standard care, and the team decides not to publish. This is problematic because:

- a. Null results are always publishable in high-impact journals
 - b. Suppressing null findings distorts the evidence base and wastes participants' contribution
 - c. The funder requires a positive result
 - d. It has no ethical implications
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Answers

1. **b.** Payment large enough to override a person's judgement about risk undermines voluntariness. Compensation should reasonably cover time and costs, not act as an incentive that makes refusal unrealistic.
 2. **b.** Because identifiers exist and the researcher can link data to individuals, this is confidentiality. Anonymity would require that no one, including the researcher, could make that link.
 3. **b.** Authorship requires substantial contribution, drafting or critical revision, final approval and accountability. Position or provision of facilities does not qualify; acknowledgement is the appropriate place.
 4. **b.** Approval cannot be retrospective. Data gathered before it lack ethical authorisation, journals require the approval date, and discovering the discrepancy after publication commonly leads to retraction.
 5. **b.** Generated references are frequently fabricated. Citing sources you have not verified and read is a fabrication of the evidentiary record, regardless of how the citation was produced.
 6. **b.** Selective publication of positive findings produces publication bias, misleading later reviews and clinical decisions. Participants accepted burden on the understanding the knowledge would be shared.
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